

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. If $A_n = \{n, n + 1, n + 2, \dots\}$, then, $\bigcap_{n=1}^{\infty} A_n = \dots$
- (A) 1. (B) $\emptyset$.
(C) ∞ . (D) n .
2. If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ which of the following is not a member of $A \times B$.
- (A) (1,4). (B) (2, 5).
(C) (3, 4). (D) (4, 3).
3. Which of the following subset of $A \times A$ defines a function on $A = \{x \in \mathbb{R} : -1 \leq x \leq 1\}$.
- (A) $C = \{(x, y) : x^2 + y^2 = 1\}$. (B) $C = \{(x, y) : x + y^2 = 1\}$.
(C) $C = \{(x, y) : x^2 + y = 1\}$. (D) None of these.
4. Which of the following set is *not* countable ?
- (A) $\{1, 2, \dots, n\}$. (B) The set $\mathbb{N}$ of natural numbers.
(C) The set $\mathbb{Q}$ of rational numbers. (D) The interval $(0, 1)$.
5. The number of injections from $S = \{1, 2\}$ to $T = \{a, b, c\}$ is ———.
- (A) 2. (B) 4.
(C) 6. (D) 8.
6. If $a \in \mathbb{R}$ such that, $0 \leq a < \epsilon$ for every $\epsilon > 0$ then, :
- (A) $a > 0$. (B) $a \neq 0$.
(C) $a = 0$. (D) None of these.
7. The binary representation of $3/8$ is, :
- (A) 0.0111111..... (B) 0.0101000.....
(C) 0.1011111..... (D) 0.0101111.....
8. If $0 < b < 1$, $\lim (b^n)$ equal to :
- (A) 0. (B) 1.
(C) b . (D) ∞ .
9. Limit of the sequence $\left(\frac{3n+2}{2n+1}\right)$ is ———.
- (A) 3. (B) $1/2$.
(C) 2. (D) $3/2$.

10. The smallest value of $K(\epsilon)$ corresponding to $\epsilon = .01$ for the sequence $\left(\frac{1}{n}\right)$ is ———.

- (A) 10. (B) 50.
(C) 100. (D) 101.

11. Which of the following is false ?

- (A) If (x_n) is a convergent sequence then $\left(x_n^2\right)$ is convergent.
(B) If (x_n) is a convergent sequence, and $x_n \geq 0$ for every n , then $(\sqrt{x_n})$ is convergent.
(C) If $\left(x_n^2\right)$ is a convergent sequence then (x_n) is convergent.
(D) If (x_n) is a convergent sequence then $\left(x_n^3\right)$ is convergent.

12. If $0 < a < b$, then $\lim \frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is :

- (A) b . (B) a .
(C) $a + b$. (D) ∞ .

13. The limit of the sequence defined inductively by, $x_1 = 1$ and $x_{n+1} = 2 + \frac{1}{x_n}$ is :

- (A) $1 - \sqrt{2}$. (B) $1 + \sqrt{2}$.
(C) $2 + \sqrt{2}$. (D) $2 - \sqrt{2}$.

14. Which of the following sequences with n^{th} term x_n diverges ?

- (A) $x_n = 1 - \frac{(-1)^n}{n}$. (B) $x_n = \frac{1 - (-1)^n}{n}$.
(C) $x_n = 1 - (-1)^n + \frac{1}{n}$. (D) $x_n = \frac{(-1)^n (n+1)}{n^2 + 1}$.

15. Which of the following statement is *not* true about closed sets ?

- (A) Arbitrary union of closed sets is closed.
- (B) Arbitrary intersection of closed sets is closed.
- (C) If $X = (x_n)$ is a sequence of elements in a closed set F , then $\lim X$ belongs to F .
- (D) A subset of $\mathbb{R}$ is closed if and only if it contains all of its cluster points.

16. Which of the following statements is *not* true about Cantor set ?

- (A) Cantor set is closed.
- (B) Cantor set is uncountable..
- (C) The complement of Cantor set in $[0, 1]$ has length 1.
- (D) Cantor set has non-empty open intervals as subsets.

17. If $z = (x, y)$ is a complex number its inverse z^{-1} is :

- (A) $\left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right)$
- (B) $\left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$
- (C) $\left(\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$
- (D) $\left(\frac{y}{x^2 + y^2}, \frac{-x}{x^2 + y^2} \right)$

18. $(1 - i)^4$ is equal to .

- (A) 4.
- (B) $4i$.
- (C) -4.
- (D) $-4i$.

19. If $z = \frac{-2}{1 + \sqrt{3}i}$, then $\text{Arg } z$ is :

- (A) $\frac{\pi}{3}$.
- (B) $-\frac{\pi}{3}$.
- (C) $\frac{2\pi}{3}$.
- (D) $-\frac{2\pi}{3}$.

20. $|e^{i\theta}|$ is equal to :

- (A) $\sqrt{2}$.
- (B) $-\sqrt{2}$.
- (C) 1.
- (D) -1.